



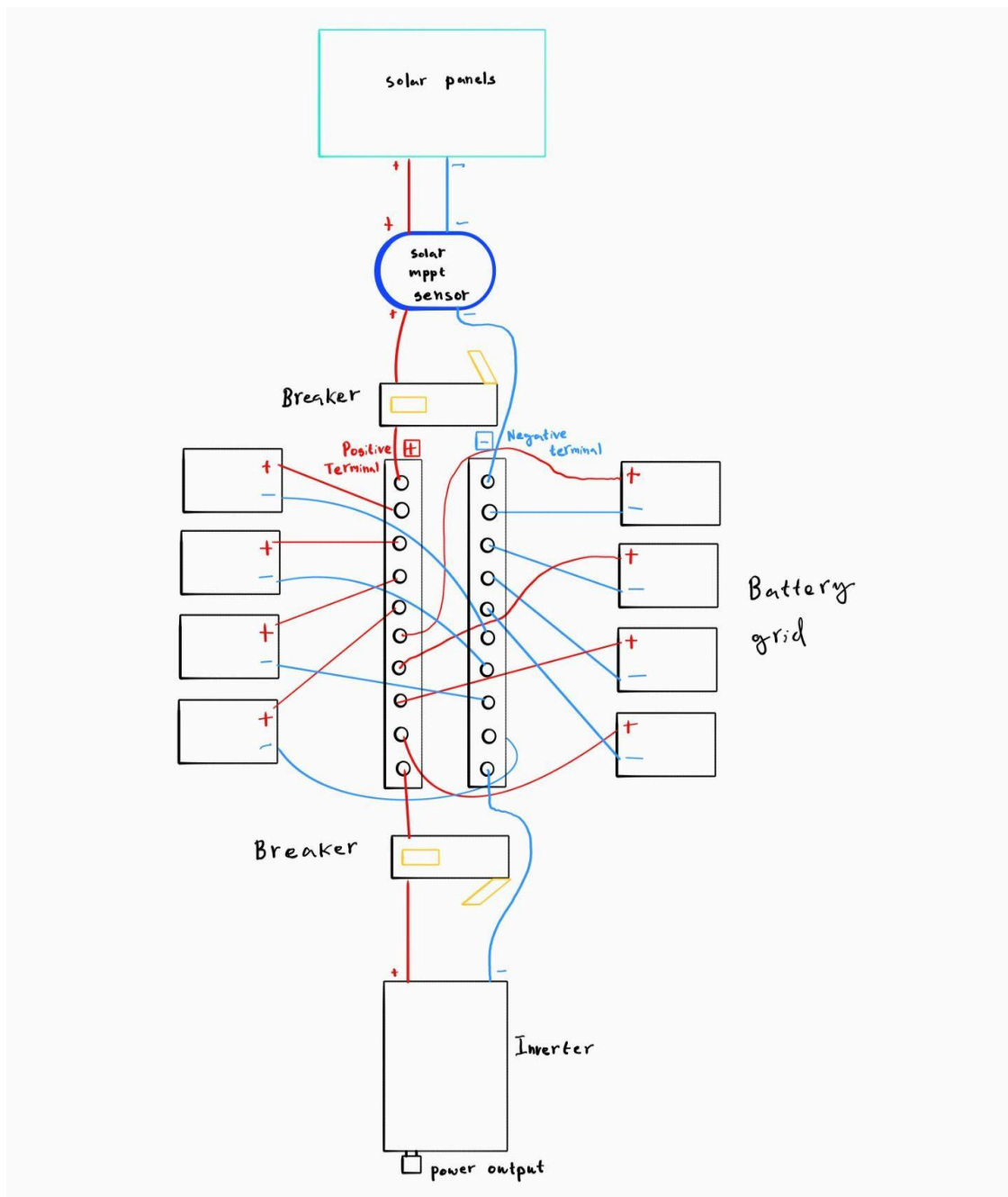
Solar Generator
User Manual
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Prepared for:
Global Strategies
San Jose, Costa Rica

Caution:

- Insulate your tools: Use rubber-handled tools to prevent accidental shorts.
- Remove metal jewelry: Watches, rings, and bracelets can cause dangerous short circuits.
- Proper Polarities: NEVER reverse positive (+) and negative (-) connections. Mark your wires clearly. Use red tape for positive and black or blue for negative.
- PPE: Wear safety glasses and electrical-rated gloves.

Overall system overview



Follow these numbered steps in order to install your system safely.

Phase 1: Solar Collection & MPPT Setup

1. **Mount Panels:** Securely mount your solar panels according to the manufacturer's orientation requirements.



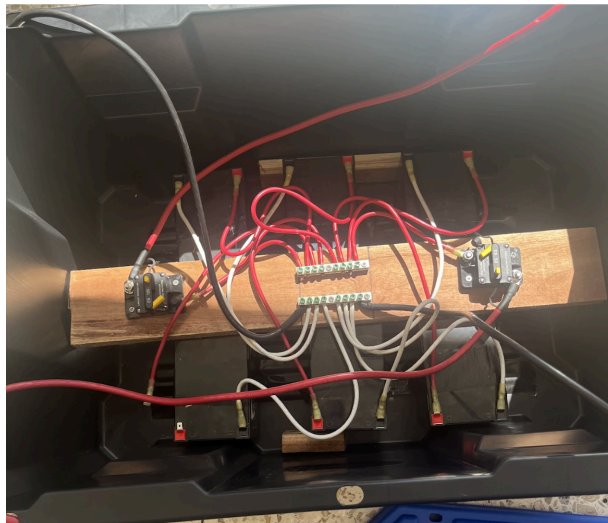
2. **Connect to MPPT:** Run the positive (red) and negative (blue) leads from your solar panels into the **Solar MPPT Sensor**.



3. **Bluetooth Link:** Power on the MPPT sensor and use the designated mobile app to pair your phone. Verify you are receiving a voltage reading from the panels.
4. **Install Input Breaker:** Connect the output leads of the MPPT sensor to the first **DC Breaker**. Keep this breaker in the **OFF** position.

Phase 2: Battery Grid & Bus Bar Assembly

5. **Mount Bus Bars:** Install your positive and negative bus bars in a well-ventilated, dry area near your battery bank.
6. **Connect to Breaker 1:** Connect the output of the input breaker to the top terminals of the bus bars (Positive to the Positive bar, Negative to the Negative bar).
7. **Parallel Battery Wiring:** * Connect every battery's positive (+) terminal to the positive bus bar.
 - Connect every battery's negative (-) terminal to the negative bus bar.
 - Ensure all connections are tightened to the torque specifications provided by the battery manufacturer.



Phase 3: Inverter & Power Output

8. **Install Output Breaker:** Connect the bottom terminals of the bus bars to the second **DC Breaker**. This protects the system from battery-side surges.

9. **Connect Inverter:** Wire the output of the second breaker to the DC input



terminals of your inverter. **Triple-check polarity:** Connect Positive (+) to Positive and Negative (-) to Negative.

10. **Final Inspection:** Perform a "tug test" on every wire to ensure no terminals are loose. Loose connections will arc and create a fire hazard.

Phase 4: Commissioning the System

11. **Verify Voltage:** Use a digital multimeter to check the voltage at the inverter input. Ensure it matches the expected battery bank voltage.
12. **Enable Solar Input:** Turn **Breaker 1** to the **ON** position. Monitor the app to ensure the batteries are beginning to receive charge.
13. **Enable Inverter:** Turn **Breaker 2** to the **ON** position.
14. **System Start:** Turn on your inverter and verify the AC output (power output) is active.

Maintenance Schedule

- **Monthly:** Visually inspect all terminals for signs of heat discoloration. Tighten if loose.
- **Quarterly:** Clean solar panels and check your MPPT app for any logged battery health errors.
- **Annually:** Verify the integrity of all wiring insulation and ensure no moisture has entered the enclosures.

